

Local Separation Is Enough

A characteristic-polynomial upgrade to the split-before-lift attack on QR-UOV

Research-program handoff note

Candidate argument pending independent expert review

30 July 2026

Status. This note strengthens the accompanying split-before-lift candidate. A second adversarial pass corrected the infinite-branch specialization proof, added a previously omitted one-time separator-leaf charge, proved the sharpened root-probability bound explicitly, and added conservative sensitivity ledgers. The result remains model-based rather than a measured implementation claim; the characteristic-polynomial specialization, rational-filter graph, and explicit online-product schedule remain the principal points requiring external review.

Abstract

The split-before-lift attack used an auxiliary field large enough for one linear form to separate all $C = 2^a$ homotopy branches, paying a $C^2/|K'|$ bad-event bound. We show that global separation is unnecessary. The characteristic eliminant and its scalar interpolation polars are obtained from the generic characteristic polynomial of the saturated homotopy algebra and retain degree at most C_0 even when the chosen linear form has collisions. To recover a forgery it suffices that the linear form have the correct valuation on branches at infinity and separate one canonical eligible target point from the other finite limits, at most $C - 1$ bad hyperplanes in total.

On a simple target factor, one Frobenius sketch using the separator itself is exact: a non-base-field point passing the test would collide with its Frobenius conjugate, contradicting simplicity. Thus the second linear numerator, the probabilistic false-positive term, the root cap, and all but one coordinate re-lift disappear. Retuning the extension degree gives a Level-I stress estimate of $2^{140.383995}$ Boolean gates, 2.616 bits below the 2^{143} target, under the inherited accounting model. Replacing the sharpened root bound by the parent manuscript’s already-proved bound still gives $2^{140.531461}$. The previous split-before-lift estimate was $2^{141.125734}$.

1 Headline

Let $q = 127$, $a = m - 4$, $C = 2^a$, and $C_0 = 2^{a-1}(a + 2)$ as in the parent attack. The supplied audit optimizes the extension degree over certified sparse irreducible polynomials.

Theorem 1.1 (Candidate local-separator bound). *Assume the public-random model, homotopy degree bound, and Boolean-gate conventions of the parent manuscript and the split-before-lift note. Replace global separation by the local conditions in section 3, accept only simple linear target factors,*

and use the exact descent test in section 4. Then the retry-aware diagnostic costs are those in table 1. In particular, Level I costs

$$2^{140.383995063}$$

Boolean gates, leaving 2.616004937 bits below the category target.

Table 1: Audited stress estimates. “Global split” is the preceding split-before-lift bundle.

Level	Parent	Global split	Local split	Target	Margin
I	151.639	141.126	140.384	143	+2.616
III	203.507	191.630	191.027	207	+15.973
V	260.389	247.637	247.125	272	+24.875

The Level-I optimum in the main ledger uses $K' = \mathbb{F}_{1278}$ with the Rabin-certified modulus

$$X^8 + X^4 - 1.$$

Degree 9, using $X^9 + X^2 + 1$, gives total 140.418965 and is retained as a nearby cross-check.

2 Characteristic polynomials do not need separation

Let $\Gamma \subset \mathbb{A}^{1+a}$ be the union of the saturated homotopy-curve components that dominate the t -line, with degree at most $D = C_0$. The Jacobian saturation removes the non-etale locus, so over $K'(t)$ its generic fiber is a finite product of separable field extensions,

$$B = K'(t)[X_1, \dots, X_a]/J, \quad \dim_{K'(t)} B = C.$$

Write x_j for the image of X_j and M_{x_j} for multiplication by x_j on B .

Introduce independent coefficients $\Lambda = (\Lambda_1, \dots, \Lambda_a)$ and the generic linear element

$$u_\Lambda = \sum_j \Lambda_j x_j.$$

Its characteristic polynomial is

$$\chi_\Lambda(t, U) = \det \left(UI - \sum_j \Lambda_j M_{x_j} \right). \quad (1)$$

Schost proves two facts that are exactly suited to the present use: specialization of Λ gives the characteristic polynomial of the specialized linear element, and the generic characteristic polynomial can be written

$$\chi_\Lambda = \frac{\Xi_\Lambda}{\Theta},$$

where the t -degrees of Ξ_Λ and Θ are at most $\deg \Gamma \leq D$ and Θ is independent of Λ [2, Lemmas 2–3 and Proposition 2]. This is the characteristic-polynomial, or generic u -resultant, form of the parametric degree theorem.

Schost phrases one corollary using a primitive element with coefficients in the ground field, together with a field-size hypothesis. The generic characteristic polynomial itself does not require such an

element. After extending to an algebraic closure of $K'(t)$, the finite etale algebra has C distinct geometric points. For every distinct pair, the difference

$$\Lambda \cdot (x_s - x_r)$$

is a nonzero polynomial in the independent indeterminates Λ . Hence u_Λ separates the geometric points over the rational-function field $K'(t)(\Lambda)$ and is primitive there, irrespective of the size of K' . Apply Schost's degree argument after this transcendental scalar extension. The determinant in (1) is formed from the original multiplication matrices, so its coefficients already lie in $K'(t)[\Lambda, U]$; scalar extension cannot lower a reduced t -degree and thereby hide a larger degree over K' . Thus no K' -rational global separator, and in particular no $|K'| \gg C^2$ assumption, is used.

Lemma 2.1 (Nonseparating eliminant). *For every $\lambda \in K'^a$, including a colliding one,*

$$q_\lambda(t, U) := \prod_{s=1}^C (U - \lambda(x_s(t))) = \chi_\Lambda(t, U)|_{\Lambda=\lambda}.$$

Every coefficient of q_λ has a common-denominator representation with numerator and denominator t -degrees at most D .

Proof. The first identity is the characteristic-polynomial product formula in the reduced generic fiber, with repeated factors allowed after specialization. The second assertion follows by substituting $\Lambda = \lambda$ in Ξ_Λ/Θ . Since Θ is independent of Λ , a separator collision cannot create a new parameter denominator or increase its degree. $\square$

The coordinate interpolation numerators are already encoded in the same generic characteristic polynomial.

Lemma 2.2 (Characteristic-polynomial polar). *Define*

$$v_j(t, U) = \sum_s x_{s,j}(t) \prod_{r \neq s} (U - \lambda(x_r(t))).$$

Then

$$v_j(t, U) = - \left. \frac{\partial \chi_\Lambda(t, U)}{\partial \Lambda_j} \right|_{\Lambda=\lambda}. \quad (2)$$

Consequently each v_j , and every fixed affine-linear combination of the v_j and q_U , has numerator and denominator t -degrees at most D , without assuming that λ separates the branches.

Proof. Over a splitting field, differentiate

$$\chi_\Lambda(t, U) = \prod_s \left(U - \sum_i \Lambda_i x_{s,i}(t) \right)$$

with respect to Λ_j and then specialize $\Lambda = \lambda$. This gives (2). Differentiating Ξ_Λ in the linear-form coefficients does not increase its t -degree, while the denominator Θ is unchanged. $\square$

This directly covers the sole linear sketch used later:

$$b_1(P) = \lambda^{q^{d-1}} \cdot P.$$

Its interpolation numerator is the corresponding fixed K' -linear combination of the v_j .

For the rational oil filter

$$\psi(X) = \frac{H_0(X)}{c + \lambda(X)},$$

the preceding split-before-lift note proves that the saturated graph curve has degree at most $2D$. Apply the same construction to its generic linear element

$$u_{\Lambda, \mu} = \Lambda \cdot X + \mu Y,$$

where $Y = \psi(X)$ is the graph coordinate. Specializing $(\Lambda, \mu) = (\lambda, 0)$ gives q_λ , and

$$-\left. \frac{\partial \chi_{\Lambda, \mu}}{\partial \mu} \right|_{(\lambda, 0)} = v_\psi.$$

The graph degree bound therefore gives numerator and denominator t -degrees at most $2D$ for the filter numerator. Consequently the existing precision

$$P = 4C_0 + 1$$

remains valid.

Remark 2.3. Separation is needed only when one wants the linear element to be a primitive coordinate for every point. The attack instead keeps the full characteristic polynomial and later accepts only a simple target factor. Characteristic polynomials and their coefficient polars remain defined at colliding linear forms.

3 Only local separation is needed

At target specialization, normalize q and each scalar numerator by the common t -valuation exactly as in Safey El Din–Schost Lemma 12 [1]. Call λ *valuation-generic* if every branch escaping to infinity with minimum coordinate valuation $\mu_s < 0$ satisfies

$$\text{ord } \lambda(x_s) = \mu_s.$$

This condition, not pairwise branch separation, is what the specialization proof uses to remove the infinite factors from the specialized eliminant.

Lemma 3.1 (Simple-factor recovery). *Assume λ is valuation-generic. Let $\bar{q}(U)$ and $\bar{v}_b(U)$ be the normalized target specializations. If $\tau \in K'$ is a simple root of $\bar{q}$, then there is exactly one finite represented target point P with $\lambda(P) = \tau$, and*

$$b(P) = \frac{\bar{v}_b(\tau)}{\bar{q}'(\tau)}.$$

No separation condition is needed away from that one fiber value.

Proof. Let S_∞ and S_0 index the escaping and finite branches. With

$$e = - \sum_{s \in S_\infty} \mu_s,$$

valuation genericity gives

$$(t^e q)(0, U) = \gamma \prod_{s \in S_0} (U - \lambda(P_s))$$

for a nonzero scalar γ . In the normalized numerator, the summand indexed by a finite branch s specializes to

$$\gamma b(P_s) \prod_{r \in S_0, r \neq s} (U - \lambda(P_r)).$$

A summand indexed by an escaping branch need not vanish term by term; rather, every such surviving summand is divisible by the full finite product $\prod_{r \in S_0} (U - \lambda(P_r))$. It therefore vanishes modulo $\bar{q}$, and in particular at every finite root of $\bar{q}$. Thus $\bar{v}_b$ modulo $\bar{q}$ is the ordinary interpolation numerator on the finite limits. A simple root τ belongs to exactly one finite branch, and evaluation followed by division by $\bar{q}'(\tau)$ gives the claimed value. $\square$

Condition on the existence of a canonical eligible nonsingular base-field target point $P_\star$; for example, choose the first under any fixed ordering. Nonsingularity makes it an isolated multiplicity-one point. Let I generic branches escape to infinity and let F have finite target limits, counted with branch multiplicity. Then $I + F = C$. There are at most I valuation-cancellation hyperplanes and at most $F - 1$ collision hyperplanes against $P_\star$. Each bad equality

$$\lambda(w) = 0 \quad \text{or} \quad \lambda(P - P_\star) = 0$$

cuts out at most $|K'|^{a-1}$ choices of λ in $(K')^a$. Thus a uniform $\lambda \in (K')^a$ fails valuation genericity or collides at $P_\star$ with probability at most

$$\frac{I + F - 1}{|K'|} = \frac{C - 1}{|K'|}. \quad (3)$$

The coefficient vector of the equation may lie only over an algebraic closure, so its solution set need not literally be a K' -defined hyperplane. The count is still valid: choose a nonzero coordinate of that vector and fix all other coefficients of λ ; the remaining equation has at most one solution in K' (and may have none).

For the filter denominator, first compute the C known start values $\lambda(\xi_s)$ and sample c uniformly from the complement of the forbidden set $\{-\lambda(\xi_s)\}_s$ in K' . Rejection sampling costs an expected factor at most $(1 - C/|K'|)^{-1}$ for a scan of the start values; this scan is negligible beside one length- P filter lift and is included in the miscellaneous envelope. Conditional on this choice, only the at most $F \leq C$ finite target values remain forbidden. Therefore

$$\Pr[c + \lambda(P) = 0 \text{ for a finite target branch}] \leq \frac{C}{|K'| - C}. \quad (4)$$

The old $C^2/|K'|$ event has disappeared.

4 Exact one-sketch Frobenius descent

Set

$$b_1(P) = \lambda^{q^{d-1}} \cdot P, \quad K' = \mathbb{F}_{q^d}.$$

The separator value $\tau = \lambda(P)$ is already available from the factor of $\bar{q}$, so no numerator for $\lambda \cdot P$ is needed.

Lemma 4.1 (Simple-root descent is exact). *Let $\tau \in K'$ be a simple linear factor of the target eliminant, and let P be its unique represented point. Then*

$$b_1(P)^q = \tau \iff P \in \mathbb{F}_q^a.$$

Proof. The forward identity for a base-field point is immediate. Conversely,

$$b_1(P)^q = \lambda \cdot P^q.$$

If it equals $\tau = \lambda \cdot P$, then P and P^q have the same separator value. The target equations and homotopy are defined over $\mathbb{F}_q$, so the represented finite target limits are stable under Frobenius. If $P^q \neq P$, the two distinct represented points produce multiplicity at least two at $U = \tau$, contradicting simplicity. Hence $P^q = P$. $\square$

This eliminates the probabilistic $C/|K'|$ false-positive term. It also eliminates the abort cap: every simple factor passing the Frobenius test and the nonzero-square oil filter is a genuine base-field target solution. The algorithm reconstructs coordinates for the first such factor and stops, so exactly one coordinate re-lift is charged per successful forgery.

5 The sharpened root bound is explicit

For completeness, this pass expands the probability calculation inherited from the split-before-lift note. Let M count all off-oil affine roots of the normalized residual system, and let N count those with invertible affine Jacobian. The parent manuscript proves, for an actual parameter $\mu \geq 1 - q^{-1}$,

$$\mathbb{E}[M] = \mu, \quad \mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}, \quad \mathbb{E}[N] = \mu\pi,$$

where $\pi = \prod_{j=1}^{r-1} (1 - q^{-j})$. The roots occur in pairs $z, -z$, and each nonsingular normalized pair yields one eligible projective point. Applying the second Bonferroni inequality to the projective-point events and bounding pairs of nonsingular roots by pairs of all roots gives

$$\begin{aligned} \Pr[\text{eligible nonsingular projective root}] &\geq \frac{\mathbb{E}[N]}{2} - \mathbb{E}\left[\binom{M/2}{2}\right] \\ &= \frac{\mu\pi}{2} - \frac{\mu^2}{8} + \frac{(q-1)\mu q^{-r}}{8}. \end{aligned}$$

The right-hand side is increasing for $\mu \in [1 - q^{-1}, 1]$, so substituting $\mu = 1 - q^{-1}$ yields

$$p_{\text{root}} = 0.3690869822 \dots$$

As a separate fallback, the ledger also substitutes the parent manuscript's already-proved lower bound $0.326336998767509 \dots$; the Level-I total then becomes 140.531461, still 2.469 bits below target.

6 Finite ledger

At Level I the local variant has two scalar families, b_1 and ψ , instead of three. The terminal product tree therefore uses

$$\frac{a}{2}(1 + 2 \cdot 2) = 125$$

long-product units. Rational reconstruction contributes 99.885 units. Candidate coordinate evaluation is charged once over the complete Las Vegas execution, rather than per trial, and the 64-unit miscellaneous cushion is retained. This pass also charges the one-time formation of $u_s = \lambda \cdot x_s$ during that coordinate rerun.

Table 2: Level-I component ledger for extension degree $d = 8$.

Component	$\log_2$ gates
Initial base-field branch lifts	139.358
Valuation-safe filter series	132.120
Terminal scalar recombination	138.791
One on-demand coordinate re-lift	137.860
Extension-valued leaf formation	127.610
Coordinate-rerun separator leaves	124.898
Factorization and exact verification envelope	115.707
Preprocessing	63.365
Total	140.383995

The one-trial success lower bound is 0.353997, giving restart factor 2.824886. At $d = 8$, the local-separator failure bound in (3) is 0.016637, and the conditional target-denominator bound in (4) is 0.016918.

Table 3: Level-I sensitivity to the multiplier on $M(n) \log n$ in rational reconstruction, retaining $d = 8$.

Multiplier	8	10	16	32	64	128
Total	140.384	140.424	140.538	140.805	141.225	141.815

Table 4: Level-I cross-checks. Each row reoptimizes the extension degree.

Online-envelope multiplier	1	2	4	8	16
Best extension degree	8	8	9	9	9
Total	140.384	141.117	141.952	142.856	143.806

Thus the candidate remains below 2^{143} throughout the rational-reconstruction sensitivity range. It also remains below target after multiplying the entire claimed online-product envelope by eight, but not by sixteen; the finite margin therefore does not excuse the remaining obligation to provide a fully explicit schedule. The aggregate Level-I margin corresponds to an overall multiplicative cushion of $2^{2.616} = 6.13$ in the inherited gate model; this does not address memory traffic or a change of model.

7 Validation and remaining risk

The bundle includes:

1. `QRUOV_local_separator_audit.py`, which computes the optimized degree, finite probabilities, and all component costs;
2. `verify_local_separator_ledger.py`, a separately written implementation importing neither the primary audit nor its parent modules, matching all displayed totals to below $5 \cdot 10^{-10}$ bits;
3. Rabin certificates for every newly used sparse modulus; and
4. the original split-before-lift toys validating branch lifting, symmetric recombination, and Frobenius arithmetic, together with an exhaustive $\mathbb{F}_9/\mathbb{F}_3$ toy showing that a non-base point passes the one-sketch test exactly on separator collisions;
5. `toy_infinite_branch_specialization.py`, which exhibits the subtle case in which an escaping-branch numerator term survives normalization but is a multiple of the finite eliminant and hence vanishes at every finite simple root.

The principal remaining proof obligations are external checks that the characteristic-polynomial degree theorem in [theorems 2.1](#) and [2.2](#) applies to the exact saturated dominant homotopy algebra and to the rational-filter graph with the stated degree bounds. The determinant identities are exact and no defect is currently known, but this identification with the parent solver’s saturated dominant algebra is what the degree-8 retuning depends on. A second independent obligation is to turn the dyadic online-product envelope into a line-by-line finite schedule or prototype; the new multiplier table shows exactly how much error that part can absorb. The earlier caveats about the idealized public-random model, seeded transfer, and astronomical memory traffic are unchanged.

8 Assessment

This is a real strengthening rather than a cosmetic ledger improvement. The preceding candidate crossed Level I by splitting before lifting; the present argument removes the largest remaining auxiliary-field overkill. It replaces a global primitive-element event with a one-sided local isolation event and turns Frobenius filtering from probabilistic to exact. Under the inherited model, the cumulative improvement over the strict parent bundle is 11.254751 bits.

The result is now robust to far larger rational-reconstruction constants than the original 1.87-bit-margin version. It should be integrated into the main draft only after a specialist in computational algebra verifies that the characteristic-polynomial degree theorem applies to the exact saturated dominant algebra and its rational-filter graph, but it is the strongest credible route found in the program so far.

References

- [1] M. Safey El Din and E. Schost. Bit complexity for multi-homogeneous polynomial system solving—application to polynomial minimization. *Journal of Symbolic Computation*, 87:176–206, 2018. doi:10.1016/j.jsc.2017.08.001; arXiv:1605.07433.
- [2] E. Schost. Computing parametric geometric resolutions. *Applicable Algebra in Engineering, Communication and Computing*, 13(5):349–393, 2003. doi:10.1007/s00200-002-0109-x.